

Jiayi Guo

Ph.D. Student in Mathematics, Actuarial and Risk Management Sciences
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EDUCATION

University of Illinois Urbana-Champaign

Ph.D. in Mathematics, concentration in Actuarial and Risk Management Sciences | 2024-present

- Advisor: Professor Zhiyu Quan.

University of Illinois Urbana-Champaign

M.S. in Predictive Analytics and Risk Management | 2022-2023

Dalian Maritime University

B.S. in Statistics | 2018-2022

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Supervisor - Illinois Risk Lab

University of Illinois Urbana-Champaign | 2024-present

- Supervise Climate Data Innovation and Consolidation projects in collaboration with the Society of Actuaries.

Graduate Research and Teaching Assistant

Department of Actuarial and Risk Management Sciences, UIUC | 2024-present

PUBLICATIONS AND WORKING PAPER

Entity-Specific Cyber Risk Assessment Using Insurtech-Empowered Risk Factors

Jiayi Guo, Linfeng Zhang, and Zhiyu Quan | Variance, Vol. 19 (2026)

- doi:10.66573/001c.155715

Fuzzy Rule-Based Models via Space Partition and Information Granulation

Yunhui Pang, Lidong Wang, Yifei Liu, and Jiayi Guo | Neural Computing and Applications 34(19) (2022)

- doi:10.1007/s00521-022-06974-3

Starting Off on the Wrong Foot: Pitfalls in Data Preparation

Jiayi Guo, Panyi Dong, and Zhiyu Quan | Submitted for review (2026)

- arXiv:2603.18190

AWARDS

PARM Summer Research Fellow

University of Illinois Urbana-Champaign | 2026

PARM Summer Research Fellow

University of Illinois Urbana-Champaign | 2025

SELECTED PRESENTATIONS

Starting Off on the Wrong Foot: Pitfalls in Data Preparation

29th International Congress on Insurance: Mathematics and Economics | Seoul, South Korea | 2026

Entity-Specific Cyber Risk Assessment Using Insurtech-Empowered Risk Factors

60th Actuarial Research Conference | Toronto, Canada | 2025

Estimating the Climate-Change Impact on a Fixed-Income Investment Portfolio due to Physical and Transitional Risks

58th Actuarial Research Conference | Des Moines, Iowa | 2023

Active Learning Method Based on Axiomatic Fuzzy Sets and Cost-Sensitive Classification

International Conference on Neural Computing for Advanced Applications | Guangzhou, China | 2021

TEACHING

ASRM 555 - Advanced Predictive Analytics

Spring 2025 and Spring 2026

- Course development and grading.

ASRM 469 - Casualty Actuarial Mathematics

Spring 2025 and Spring 2026

- Course development, grading, and office hours.

ASRM 455 - Predictive Analytics

Fall 2024 and Fall 2025

- Course development, grading, and office hours.

PROFESSIONAL EXAMS

Society of Actuaries

Exam P; FAM (UEC); FM (UEC)

RESEARCH FOCUS

Climate and catastrophe risk; cyber risk and cyber insurance; insurance loss analytics; machine learning and physics-informed AI; predictive analytics and data preparation.